PEER REVIEW HISTORY

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ARTICLE DETAILS

TITLE (PROVISIONAL)	Developing a profile of activities of daily living for bipolar disorder: A systematic review protocol and meta-analysis
AUTHORS	Aniserowicz, Anna M.; Safi, Fatima; Colquhoun, Heather; Stier, J.; Nowrouzi-Kia, Behdin

VERSION 1 – REVIEW

REVIEWER	Matsumoto, Kazuki Chiba University
REVIEW RETURNED	02-Dec-2021

GENERAL COMMENTS	This papar is a protocol for the first systematic review of ADL in patients with bipolar disorder. The systematic review process is well designed at some stages to answer objectives of this study. It seems to be an effective protocol paper, except for concern. One minor comment of mine is related to the fact that this paper does not provide description to answer research question (2) at "Data synthesis". The result of this study is the publication of a successful systematic review, which, as the authors claim, will provide useful information to healthcare professionals, parties and their families who support bipolar disorder. Comment The authors may consider adding a method of data integration and description corresponding to the research question (2) at "Data synthesis" to make the protocol paper more effective.

REVIEWER	Chang, Wing Chung’ Univ Hong Kong, Department of Psychiatry
REVIEW RETURNED	14-Dec-2021

GENERAL COMMENTS	This is a research protocol of a systematic review which aims to examine and summarize ADL profile among bipolar disorder patients, in particular the level of independence for specific ADLs. The study will adopt systematic review but may not go for meta-analysis. There are some important methodological issues that merit further elaboration or clarification: 1. Why meta-analysis is not included in the protocol in the first place? The authors stated in Data synthesis section that "if appropriate, a meta-analysis will be used to group...". However, no details on statistical tests about meta-analysis were provided. 2. The review did not specify in detail how to further classify the patient population. For example, by diagnosis, does it mean that the review will subdivide the population into bipolar I, II and others for subgroup analysis? For symptoms, do they mean that the review will subdivide BD into different phases e.g. euthymic,

	depressed, manic states for subgroup analysis. This is critically important. Given that ADL and work and social functioning are expected to be severely impaired when patients are in acutely manic or depressed state, versus euthymic state. And this is the euthymic state that is most important in determining the degree of impairment in ADL associated with BD patients. 3. Another important aspect is the need to consider whether patient samples assessed are chronically ill or in early stage of the illness. It is expected that ADL impairment would likely be more prominent and persistent in chronic relative to early illness stage. 4. How to define cognitive subgroup in patient population (for symptoms)? Further detail is required. This could not be directly compared with manic, depressed and euthymic states.

VERSION 1 – AUTHOR RESPONSE

Reviewer: 1
Dr. Kazuki Matsumoto, Chiba University
Comments to the Author:
This papar is a protocol for the first systematic review of ADL in patients with bipolar disorder. The systematic review process is well designed at some stages to aswer objectives of this study. It seems to be an effective protocol paper, except for concern. One minor comment of mine is related to the fact that this paper does not provide description to answer research question (2) at "Data synthesis". The result of this study is the publication of a successful systematic review, which, as the authors claim, will provide useful information to healthcare professionals, parties and their families who support bipolar disorder.

Comment
The authors may consider adding a method of data integration and description corresponding to the research question (2) at "Data synthesis" to make the protocol paper more effective.

R3: Thank you for the suggestion. In the revised copy, we have updated question 2 to focus on difficulties with ADL functioning (page 7, lines 1-2). In the methods section, we have included additional information on how the data will be integrated for the analysis. To address research question (2), our plan is to complete both a qualitative, descriptive analysis, and if there is sufficient data, our plan is to complete a meta-analysis (page 12).

The descriptive analysis would involve categorizing the different ADLs studied in primary literature. We plan to group by similar ADLs, including self-care, productivity, and leisure. We will qualitatively describe general findings across the studies identified and draw inferences on which ADLs patients have most difficulties with functioning. The specific changes to the manuscript be found on page 12, lines 3-10. We also plan to complete a meta-analysis as the quantitative component to compare the effects where ADL assessments or statistical procedures are similar. More information regarding the meta-analysis can be found on page 12, lines 11-19.

Reviewer: 2
Dr. Wing Chung Chang, Univ Hong Kong
Comments to the Author:
This is a research protocol of a systematic review which aims to examine and summarize ADL profile among bipolar disorder patients, in particular the level of independence for specific ADLs. The study will adopt systematic review but may not go for meta-analysis. There are some important methodological issues that merit further elaboration or clarification:

1. Why meta-analysis is not included in the protocol in the first place? The authors stated in Data synthesis section that "if appropriate, a meta-analysis will be used to group...". However, no details on statistical tests about meta-analysis were provided.

R4: We have revised the data synthesis section to include the steps that will be taken for a meta-analysis. A meta-analysis will be included if there is sufficient statistical information. For example, if we found a common assessment that was used across multiple studies, we plan to conduct meta-analyses to calculate overall effect sizes. If statistical procedures are found that are comparable across studies (for example, relative risks or odds ratio), we plan to compare them using forest plots. Statistical analysis will be completed using R programming. Additional information about the meta-analysis can be located on page 12, lines 11-19.

2. The review did not specify in detail how to further classify the patient population. For example, by diagnosis, does it mean that the review will subdivide the population into bipolar I, II and others for subgroup analysis? For symptoms, do they mean that the review will subdivide BD into different phases e.g. euthymic, depressed, manic states for subgroup analysis. This is critically important. Given that ADL and work and social functioning are expected to be severely impaired when patients are in acutely manic or depressed state, versus euthymic state. And this is the euthymic state that is most important in determining the degree of impairment in ADL associated with BD patients.

R5: Thank you for bringing this to our attention. We have revised the protocol to include information regarding the phases in the summary of study findings, if applicable to the study. This revision is located on page 10, line 13. We also included how we will subdivide the population for data synthesis in the “ADL Profile” table – please see the end of page 10 and page 11, lines 1-2. Curently, many primary literature articles tend to focus on bipolar disorder without separating between episodes. Therefore, there may not be sufficient data to develop this information.

3. Another important aspect is the need to consider whether patient samples assessed are chronically ill or in early stage of the illness. It is expected that ADL impairment would likely be more prominent and persistent in chronic relative to early illness stage.

R6: Thank you for raising this important point. We agree that age of onset and duration of illness would be very helpful in identifying patterns for functioning. We would also infer that if a population had an earlier onset or longer duration of illness, there would be more pronounced challenges in the ADL functioning.

From our scan of the literature, most current literature on bipolar disorder and ADL functioning typically does not report subgroups the level of ADL functioning based on illness progression or age of onset. This may limit the conclusions from the studies at the individual level. During an informal scan of the literature, we have noticed that many studies provide the average duration and age of onset for their sample. To address onset and duration of illness, the data synthesis section, Table 1 will also include each study’s reported average duration of illness of the bipolar disorder patients if it is included in the original study (page 10, line 13-14).

4. How to define cognitive subgroup in patient population (for symptoms)? Further detail is required. This could not be directly compared with manic, depressed and euthymic states.

R7: Thank you for the suggestion. Once we have conducted our formal searches, and if there are a sufficient number of studies that examine the subtypes as suggested, we will examine them for patterns. The studies are quite heterogeneous, whether in the symptom severity or the ADL assessments used. We have found studies that look at the different subtypes, but most studies only

look at bipolar disorder as a whole. Some amendments have been made in the inclusion criteria (page 8, lines 11-12).

We want to thank all the reviewers for their suggestions. We believe their feedback has improved the quality of the paper.

VERSION 2 – REVIEW

REVIEWER	Chang, Wing Chung Univ Hong Kong, Department of Psychiatry
REVIEW RETURNED	12-Mar-2022

GENERAL COMMENTS	The authors have adequately addressed reviewers' comments. I have no further comment for the revised manuscript.